Pankhuri Vanjani

Doctoral Researcher, Intuitive Robots Lab · Karlsruhe Institute of Technology, Germany

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Education

Karlsruhe Institute of Technology, Intuitive Robots Lab — *Karlsruhe, Germany*

Ph.D. in Computer Science — Robot Learning — *Oct 2023 - present*

- Advisor: Prof. Rudolf Lioutikov. Developing multimodal robot foundation models, including vision-language-action and world action models, for precise manipulation. My work explores multimodal representations, pretraining and model adaptation to build policies that are fast, reactive and robust in real-world environments.

Saarland University — *Saarbrücken, Germany*

M.Sc. in Embedded Systems (Computer Science) — *Oct 2020 - Sep 2023*

- Thesis: *Framework for 3D Human Motion Generation under Zero Gravity: Dataset and Benchmarking*.

The LNM Institute of Information Technology — *Jaipur, India*

B.Tech. in Electronics and Communication Engineering — *Jul 2016 - Jul 2020*

- Gold Medalist, graduating batch of 2020 (best all-round performance).
- Thesis: *Vision-Based Autonomous Navigation for a Robot in Indoor Environments*. Studied vision-based localization and path planning with a focus on SLAM, comparing Kinect, RealSense D435 and monocular cameras.

Research Experience

Max Planck Institute for Informatics — *Saarbrücken, Germany*

Student Researcher (M.Sc. Thesis), Visual Computing and AI — *Jun 2022 - Sep 2023*

- **3D human motion generation under zero gravity:** Created a physically plausible synthetic dataset of human motion in zero gravity from multi-view motion capture using physics-based simulation and optimization, and benchmarked state-of-the-art 3D human pose estimation methods on it.

German Research Center for Artificial Intelligence (DFKI) — *Saarbrücken, Germany*

Research Assistant, Augmented Vision; Agents and Simulated Reality — *Oct 2021 - Dec 2022*

- **LiDAR SLAM for autonomous driving:** Improved mapping and localization from LiDAR point clouds collected on campus using hdl_graph_slam and interactive SLAM.

National University of Singapore (NUS) — *Singapore*

Visiting Research Intern, Acoustic Research Lab, under Prof. Mandar Chitre — *Dec 2019 - Jun 2020*

- **ROS2 navigation with Unity for underwater vehicles:** Developed control and navigation software for underwater robots using ROS2 and micro-ROS.

JdeRobot, Universidad Rey Juan Carlos — *Spain (remote)*

Google Summer of Code Intern — *May 2019 - Aug 2019*

- **Open-source ROS2 migration:** Migrated five JdeRobot tools (e.g., CamViz, CarViz, VisualStates) to ROS2 and built a ROS2 communication library replacing ICE; several packages were released as official ROS packages.

Indraprastha Institute of Information Technology (IIIT-Delhi) — *Delhi, India*

Summer Research Intern, under Dr. P.B. Sujit — *May 2018 - Jul 2018*

- **Collision avoidance for cooperative multi-UAV systems:** Implemented a collision-avoidance algorithm based on the collision-cone approach for multiple UAVs in ROS and Gazebo.

Publications

Conference Papers

- **DAM-VLA: Decoupled Asynchronous Multimodal Vision Language Action Model**
  P. Vanjani, Z. Li, J. Suliga, M. Reuss, G. Geraci, X. Jiang, R. Lioutikov
  Conference on Robot Learning (CoRL), 2026.
- **DisDP: Robust Imitation Learning via Disentangled Diffusion Policies**
  P. Vanjani, P. Mattes, K. D. Kuryshev, X. Jia, V. Dave, R. Lioutikov
  Reinforcement Learning Conference (RLC), 2025.

- **SPARC: Reliable Spatial Annotations from Robot Demonstrations at Scale**
  N. Blank, P. Mattes, M. X. Li, J. Suliga, T. Roth, M. Reuss, **P. Vanjani**, R. Lioutikov
  Conference on Robot Learning (CoRL), 2026.

Submitted Manuscripts, Under Review

- **ReCAT: Remember, Count, and Time: Structured Recurrent Memory for Robot Manipulation**
  P. Vanjani, M. Hatab, C. Mızraklı, V. Shaj, Z. Li, M. Reuss, R. Lioutikov
  arXiv preprint arXiv:2609.35200 | Memory for Robot Foundation Models Workshop, CoRL 2026.
- **Touch Improves Learning Dexterous Grasping from Demonstration**
  N. Leukroth*, **P. Vanjani***, J. Suliga, K. Nakahara, R. Lioutikov, R. Calandra
  RSS 2026 Workshops: 4th Workshop on Dexterous Manipulation, **Oral**; Tactile Sensing for Robotic Foundation Models.
- **TACTIC: Understanding Tactile Encoders and Conditioning for Contact-rich Robot Manipulation Policies**
  S. Bien, D. Oliveira Makowski, C. Kneissl, R. Mirjalili, **P. Vanjani**, R. Lioutikov, G. Kutyniok, F. Walter, W. Burgard
  arXiv preprint arXiv:2609.30969 | 3rd Workshop on Dexterous Manipulation, CoRL 2026.
- **Correct Robots Before They Make Mistakes: Proactive Human-in-the-Loop Intervention via Preference**
  X. Jiang, J. Jiang, H. Zhou, **P. Vanjani**, A. Baki, J. Suliga, G. Li, Z. Li, G. Neumann, R. Lioutikov

* equal contribution

Workshop Papers

- **Few-Shot Residual Adaptation of Pretrained Latent World Models** *Supervised project*
  J. Suliga, **P. Vanjani**, N. Nilavadi, W. Burgard, R. Lioutikov
  8th Robot Learning Workshop (WRL@NeurIPS 2026), **Oral**; Continual World Models Workshop, NeurIPS 2026;
  CoRL 2026 Workshops: Robot Learning at Deployment; Do Robots Need World Models?; Continually Self-Improving Robots;
  Bringing Physics Simulation and World Models Together for Robotic Manipulation.
- **What Makes a VLA Reactive? A Study of Execution Strategies for Dynamic Manipulation** *Supervised project*
  Z. Li, Z. Han, **P. Vanjani**, W. Li, R. Lioutikov | Efficient Foundation Models for Real-Time Embodied AI Workshop, CoRL 2026.

Earlier Publications

- **Communication and networking technologies for UAVs: A survey**
  A. Sharma, **P. Vanjani**, N. Paliwal, C. M. W. Basnayaka, D. N. K. Jayakody, H. C. Wang, P. Muthuchidambaranathan
  Journal of Network and Computer Applications, 2020.
- **Image processing-based intelligent robotic system for assistance of agricultural crops**
  N. Paliwal, **P. Vanjani**, J.-W. Liu, S. Saini, A. Sharma
  International Journal of Social and Humanistic Computing, 3(2):191–204, 2019.

Honors & Awards

2021	**ETH Zurich Robotics Summer School**, **2nd place**, Search and Rescue Challenge	*Zurich*
2019-24	**Google Summer of Code, JdeRobot**, Org. Administrator (2021); Mentor (2020–24); Intern (2019)	*Spain*
2020	**Speaker, Google Mentor Summit**, Represented JdeRobot; talk on the projects I mentored	*Online*
2019	**ROSCon Diversity Scholarship**, Recipient; gave a lightning talk at the conference	*Macau*
2018	**IICDC Quarter Finalist**, India Innovation Challenge Design Contest, DST & Texas Instruments, with the *Smart Farming Bot*, a web-controlled farming-assistance robot on a BeagleBone Black.	*India*
2015	**NTSE Merit Scholarship**, National Talent Search Examination, Government of India	*India*

Supervision, Teaching & Academic Service

Student Supervision — KIT, since 2023

- **Jean Henry Maack** — B.Sc. thesis: *Natural Dialog System in German Elderly Care*.
- **Kevin Daniel Kuryshev** — B.Sc. thesis: *Context-Aware Audio Interruptions*.
- **Marc Nikolaus** — M.Sc. thesis: *Language and Audio Guided Failure Recovery Methods for Imitation Learning*.
- **Jakub Suliga** — B.Sc. thesis: *Few-Shot Residual Adaptation of Pretrained Latent World Models*.
- **Zhendong Han** — B.Sc. thesis: *Optimizing VLA Pipeline for Faster, Reactive Performance*.
- **Andreas Ott** — M.Sc. thesis: *Leveraging Temporal Scene Graphs for Long-Horizon Imitation Learning*.
- **Mattis Aydt** — B.Sc. thesis: *JEPA with a Multi-Horizon Action-Conditioned Predictor for Variable-Horizon Planning*.
- **Selim Kandemir** — M.Sc. thesis: *View-Invariant VLA Models through Explicit Conditioning on Camera Extrinsics*.
- **Ahmad Baki** — B.Sc. thesis: *Enhancing Imitation Learning with VR-Based Gaze Feedback*.

Reviewing & Teaching

- **Reviewer:** CoRL, RSS, ICRA, NeurIPS.
- **Teaching:** Robot Learning proseminar; research seminars and practical courses, KIT (since 2023).